Reduction of the Angular Amplitude Observable: the J_3 Split as Accumulated Oriented Symplectic Area

(internal note, prior to the Weil–BFS computation)

Cosmochrony programme — working note

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Abstract

This internal note fixes the numerical observable for the absolute normalisation of the angular generation split of Q14, §6 [5], before any Weil–BFS computation is written. A short Baker–Campbell–Hausdorff reduction identifies the per-step J_Π-odd J_3 component of the metaplectic cascade generator with the oriented symplectic area swept per step — the central, Carnot-degree-2 increment of Heis_3 — and not with a free shear parameter. The quantity to be measured is therefore the accumulated oriented J_3-odd area along the Weil–BFS cascade, normalised by the projected capacity $\widehat{I}(n)$; at the intrinsic saturation rank n_3^{obs} this equals $\varepsilon_{\mathrm{Weil}}(n_3^{\mathrm{obs}})$, to be compared with the dictionary value $\varepsilon = \frac{1}{10}$ *without fitting.* In the O12 construction the two sectors are the two terms of the fingerprint exponent: the frequency B_c carries the radial capacity σ_c, the central phase A_c carries the angular split. A direct evaluation on symmetric BFS shells returns $\Theta_{\mathrm{raw}} = 0$ identically: the symmetric capacity data fix the radial sector but cannot select an oriented angular branch, so the orientation prescription is an irreducible input prior to $\mathcal{N}_A$.

Keywords: metaplectic cascade; oriented symplectic area; Heisenberg grading; generation split; Weil representation; projected capacity.

1 Context

Q14, §6 [5] establishes the amplitude relation $\varepsilon = \mathcal{N}_{\mathrm{casc}}|\alpha|$ with a factorised architecture: the angular J_3 split (metaplectic, norm-preserving) and the radial capacity decay (projective, norm-changing) are Hilbert–Schmidt orthogonal generators coupled only by the cascade arrow, and $\mathcal{N}_{\mathrm{casc}}$ is a pure angular–radial transfer constant. The profile and saturation status of the amplitude are fixed; its absolute value is deferred to the physical normalisation of the metaplectic J_3 generator, the shear value $\alpha \simeq 0.156$ being a diagnostic coefficient only. This note fixes the observable that a Weil–BFS computation must measure to test that absolute value; it does not attempt to prove $\varepsilon = \frac{1}{10}$.

2 Reduction lemma

Lemma 1 (Per-step split is the oriented symplectic area)**.** *Let the cascade step be the oriented metaplectic product $g = S_X(t)\, S_Y(s)$ acting on the weight space $V = \mathbb{C}^2$ and lifted to $\mathrm{Sym}^2(V)$, with X, Y the horizontal $\mathfrak{sl}(2)$ generators and $[X, Y] = 2J_3$. Then, to second Baker–Campbell–Hausdorff order,*

$$\log\big(S_X(t)\, S_Y(s)\big) = tX + sY + \tfrac{ts}{2}\,[X, Y] + O(3),$$

and the projection onto the J_3 direction is

$$\Pi_{J_3}\,\log\bigl(S_X(t)\,S_Y(s)\bigr) = ts\,J_3 + O(3).$$

Hence the per-step J_Π-odd J_3 component is

$$\alpha_{\rm step} = ts = \textit{oriented symplectic area swept per cascade step},$$

the central (Carnot-degree-2) increment of Heis_3*, and not a free shear parameter.*

Proof. The second-order BCH expansion of $\log(e^{tX}e^{sY})$ gives the stated form, the only second-order term being $\frac{ts}{2}[X,Y]$ since X, Y are nilpotent of order two in the lift. With $[X,Y] = 2J_3$ this contributes $\frac{ts}{2}(2J_3) = ts\,J_3$, while $tX + sY$ carries no J_3 component. The dependence on t and s only through the product ts is the oriented-area form: ts is the symplectic area of the ordered step, equal to the central-coordinate increment of the Heisenberg law $(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab')$. □

Remark 1. *This is the dynamical, J_Π-odd counterpart of the static result that the covariance* $[1:\frac{1}{2}:\frac{1}{2}]$ *and its factor* 2 *originate in the Carnot-degree-2 weight of the central generator $Z = [X,Y]$ [3], in the identification with* $\mathrm{Sym}^2(V_\rho)$ *of [2]. The static factor is carried by the J_Π-even Carnot dilation; the split is carried by the J_Π-odd oriented area, a distinct direction.*

3 The Weil–BFS observable

Lemma 1 fixes what the computation must measure: not a chosen $\alpha(t,s)$, but the oriented central area actually accumulated by the Weil cascade on $\mathrm{Cay}(\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$. In the O12 construction this area is explicit. Built from the three accumulated generator steps (a_i, b_i, g_i) on the uniform seed, the fingerprint of a shell element g is a single additive character

$$\mathrm{fp}_c(g;x) = q^{-3/2}\,\exp\Bigl(\tfrac{2\pi i}{q}\bigl(A_c(g) + B_c(g)\,x\bigr)\Bigr), \qquad B_c(g) = \sum_i c_i\,b_i, \quad A_c(g) = \sum_i c_i\,(g_i - b_i a_i),$$

verified single-frequency to machine precision [6, 4]. The two terms of the exponent separate the two sectors exactly:

$$B_c \text{ (frequency)} = \text{radial / capacity}, \qquad A_c \text{ (central phase)} = \text{angular}/J_3.$$

The radial capacity is the frequency-counting functional

$$\sigma_c(n) = \frac{|\{B_c(g) : g \in S_n\}|}{|S_n|},$$

blind to A_c: scaling the whole block $c \mapsto \omega c$ permutes $B_c \mapsto \omega B_c$ bijectively and leaves σ_c invariant ([H-color]), while $A_c \mapsto \omega A_c$ carries the per-vector phase that Gram–Schmidt discards. The angular increment is therefore the J_Π-odd, oriented shell increment of the central phase A_c,

$$\Delta A^{\rm odd}_{J_3}(n) = \mathrm{Odd}_{J_\Pi}\bigl[\,\text{oriented shell increment of } A_c(g),\ g \in S_n\,\bigr],$$

where Odd_{J_Π} is the part odd under conjugation $c \mapsto q - c$ and the orientation is the cascade arrow. Define

$$\varepsilon_{\rm Weil}(n) = \sum_{m \le n} \Delta A^{\rm odd}_{J_3}(m), \qquad K^{\rm Weil}_{\rm ar}(n) = \frac{\varepsilon_{\rm Weil}(n)}{\widehat{I}(n)}, \qquad \widehat{I}(n) = \frac{\sigma_{\rm pair}(0) - \sigma_{\rm pair}(n)}{\sigma_{\rm pair}(0) - \sigma_{\rm pair}(n_{\rm sat})},$$

with $\sigma_{\mathrm{pair}} = \sigma_c\,\sigma_{q-c}$. Since $\widehat{I}(n_3^{\mathrm{obs}}) = 1$ at the intrinsic saturation rank [1], the quantity to be compared with the dictionary value is

$$K_{\mathrm{ar}}^{\mathrm{Weil}}(n_3^{\mathrm{obs}}) = \varepsilon_{\mathrm{Weil}}(n_3^{\mathrm{obs}}).$$

This eliminates the principal false positive: no shear parameters are chosen; the radial sector is the frequency count of B_c and the angular sector is the central phase A_c, separated by the form of the fingerprint itself.

4 Required outputs and honest outcomes

The Weil–BFS computation must return three quantities, not one:

1. $K_{\mathrm{ar}}^{\mathrm{Weil}}(q)$ at the saturation rank, for $q \in \{61, 101, 151\}$;
2. the profile $K_{\mathrm{ar}}^{\mathrm{Weil}}(n)$ (constant / saturating / n_3-concentrated), cross-checking the locking law of Q14, §6;
3. the sign control under cascade reversal: the J_3-odd component must flip sign, as required by the oriented arrow.

The comparison $K_{\mathrm{ar}}^{\mathrm{Weil}}(q) \to 1/10$ is a possible outcome only after an oriented branch has been selected; it must be tested, not fitted. The honest outcomes are: $K_{\mathrm{ar}}^{\mathrm{Weil}} \to 1/10$, giving an orthogonal confirmation of ε; $K_{\mathrm{ar}}^{\mathrm{Weil}} \neq 1/10$ but q-stable, meaning that the ADE dictionary fixes ε while the angular sector carries an additional transfer factor, with the common scale $\kappa = 5/12$ of the deficit dictionary a natural candidate; $K_{\mathrm{ar}}^{\mathrm{Weil}}$ q-unstable, leaving the amplitude open; or $K_{\mathrm{ar}}^{\mathrm{Weil}} = 0$ identically on symmetric BFS shells, which is the symmetric-capacity obstruction of Section 6 and means that the oriented branch is not selected by the capacity data alone.

5 The deferred normalisation $A \to \varepsilon$

The fingerprint gives a concrete angle per element, $\theta_A(g) = 2\pi A_c(g)/q$. The angular amplitude is then

$$\varepsilon_{\mathrm{Weil}} = \mathcal{N}_A \cdot \mathrm{Odd}_{J_\Pi}\left(\tfrac{2\pi A_c}{q}\right)_{\widehat{I}\text{-normalised}},$$

where $\mathcal{N}_A$ is not a fit to $\frac{1}{10}$ but the geometric factor converting a Weil central phase into the spectral coefficient of the J_3 generator on $\mathrm{Sym}^2(V_\rho)$. It is the angular analogue of the radial scale $\kappa = \frac{5}{12}$ of the deficit dictionary. This is the last open step, and it is posed only after the observable above is fixed, so that $\mathcal{N}_A$ acts on the concrete pipeline quantity θ_A rather than on an abstract area.

6 Symmetric-capacity obstruction

The preceding definition deliberately separates the radial and angular observables. The radial observable is the frequency count of B_c, while the angular observable is the central phase A_c. A direct Weil–BFS evaluation on symmetric shells gives a useful obstruction rather than an amplitude value.

On the symmetric BFS shell S_n, the involution $g \mapsto g^{-1}$ pairs opposite central phases. Consequently the oriented J_Π-odd average of the phase A_c cancels:

$$\Theta_{\mathrm{raw}}(n) = \mathrm{Odd}_{J_\Pi}\left(\frac{2\pi A_c}{q}\right)_{S_n} = 0.$$

This cancellation is exact at the shell level. It does not mean that the phase A_c is trivial; rather, it means that the symmetric capacity observable contains both orientations and therefore erases the oriented angular component.

The numerical Weil–BFS check confirms this structure for $q \in \{61, 101, 151\}$. The radial profile $\sigma_{\text{pair}}(n)$ is non-trivial and saturates through the frequency sector B_c, while the raw angular profile $\Theta_{\text{Weil}}(n)$ vanishes identically before any orientation prescription is introduced.

This is the angular analogue of the static obstruction in Q14. The symmetric capacity data determine the radial projection sector, but they do not select an oriented branch of the metaplectic cascade. Therefore the missing ingredient is not first the normalisation $\mathcal{N}_A$, but the orientation prescription itself. Only after an oriented Q11 cascade branch is specified does it make sense to ask for the geometric conversion factor

$$\varepsilon_{\text{Weil}} = \mathcal{N}_A \, \Theta_{\text{oriented}}.$$

In particular, no value of $\mathcal{N}_A$ can turn the symmetric-shell result into a non-zero generation split, since the quantity to be normalised is exactly zero.

References

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